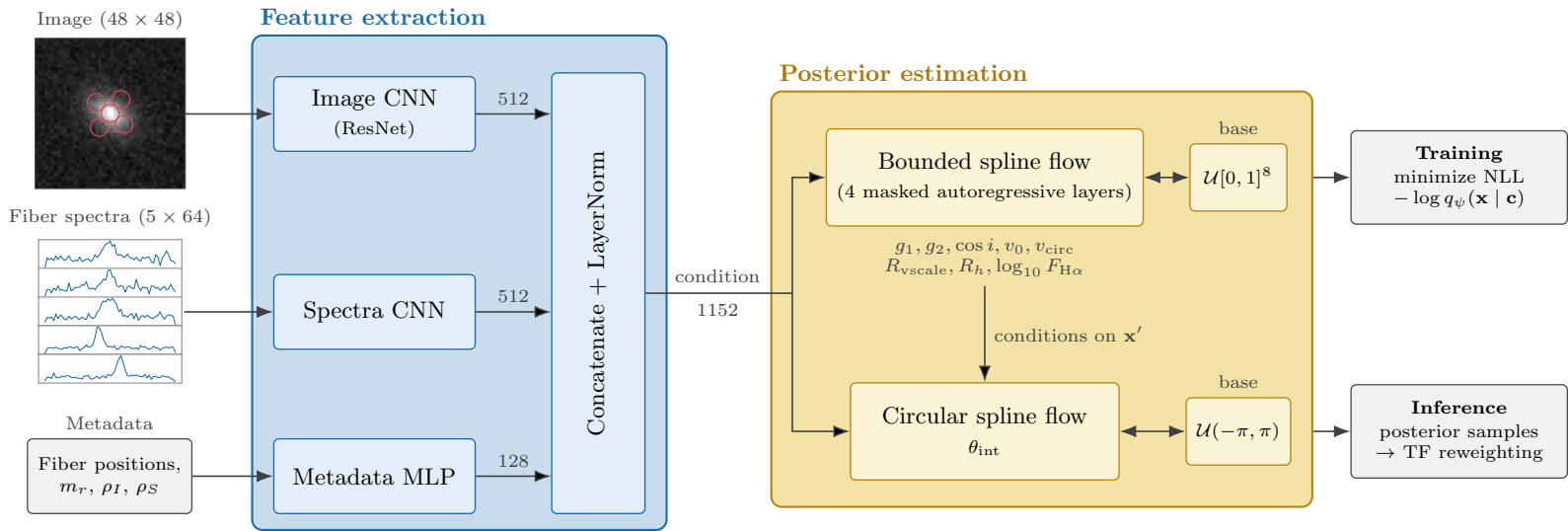

Image (48 × 48)
Feature extraction
Image CNN
(ResNet)
512
Fiber spectra (5 × 64)
Spectra CNN
512
Metadata
Fiber positions,
m_r, ρ_I, ρ_S
Metadata MLP
128
Concatenate + LayerNorm
condition
1152
Posterior estimation
Bounded spline flow
(4 masked autoregressive layers)
base
$\mathcal{U}[0,1]^8$
$g_1, g_2, \cos i, v_0, v_{\rm circ}$
$R_{\rm vscale}, R_h, \log_{10} F_{\rm H\alpha}$
conditions on $\mathbf{x}'$
Circular spline flow
$\theta_{\rm int}$
base
$\mathcal{U}(-\pi, \pi)$
Training
minimize NLL
$-\log q_\psi(\mathbf{x} \mid \mathbf{c})$
Inference
posterior samples
→ TF reweighting